



Micro-lens fabrication by a long-stroke precision stage with switching control based on model response prediction

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Abstract

This paper develops an automatic control switching mechanism for a long-stroke precision stage intended to improve micro-lens fabrication. The stage consists of a piezoelectric transducer (PZT) stage and a motor stage that achieves centimeter-length travel distances with nanometer precision. We obtained the PZT stage model by experiments and then applied robust loop-shaping techniques to design two controllers: one gave fast responses and the other gave smooth responses. The two controllers were further simplified as robust proportional-integral-derivative (PID) controllers for hardware implementation. We then developed a control switching mechanism, based on model response prediction, to combine the merits of these two controllers. For the motor stage, we derived the stage model by experiments and designed a gain-scheduling controller that regulated the stage to move at the fastest speed for long travel distances until it was within the stroke of the PZT stage. The two stages were then combined with a double-loop control structure to provide a travel range of 10 cm with a precision of 1.23 nm. We implemented the designed control mechanisms and structures for experimental verification. The combined stage was also integrated with a two-photon polymerization (TPP) system to fabricate micro-lenses. The optical properties of the micro-lenses were evaluated to demonstrate the effectiveness of the proposed control in precision positioning for long travel distances.

1 Introduction

Precision positioning is increasingly important in many industries, such as laser processing, optical inspection, and semiconductor manufacturing. Piezoelectric transducer (PZT) is often used for precision positioning because of its favorable properties (e.g., high resolution, fast responses, and large driving forces). However, the performance and applications of this material might be limited by its nonlinearities and strokes. Advanced control schemes were initially applied to solve the problems caused by PZT nonlinearities. For example, Xu and Li (2010) combined a feedforward compensator and a proportional-integral-derivative (PID) controller to produce a piezoelectric micromanipulator that attained precise positioning with a tracking errors of $\pm 0.44 \mu\text{m}$. Rakotondrabe (2011) applied a Bouc-Wen model to compensate for the

hysteresis nonlinearity in piezoelectric actuators with an inverse multiplicative scheme. Wang et al. (2016a) designed robust loop-shaping controllers for a two-dimensional PZT stage to provide strokes of $15 \mu\text{m} \times 15 \mu\text{m}$ and settling time of less than 11 ms. Subsequently, PZT stages were usually integrated with large-stroke actuators to improve the travel ranges. For instance, Zheng et al. (2011) presented a rotary dual-stage actuator consisting of a voice coil motor and a piezoelectric actuator that gave a steady-state position error of $\pm 0.06 \mu\text{m}$ for a tangential travel of 142 mm. Kwon et al. (2001) proposed a precision XY manipulator that combined a PZT stage and a motor to provide a work space of $200 \times 100 \text{ mm}$. In this paper, we combined the PZT stage with a motor stage and designed a double-loop control structure to achieve precision positioning for long travel distances.

Because a single controller cannot satisfy all the desired performance requirements, controller switching has frequently been applied to adjust controllers according to the operating conditions. For example, Zhao et al. (1993) developed a fuzzy scheme to improve system performance by adjusting the PID parameters based on error signals. Yamaguchi et al. (1996) presented a mode-switching

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control with initial value compensation that improved the positioning responses of magnetic disk drives. Wang et al. (2017) applied gain-scheduling integral control to a PZT stage to achieve nano-positioning with a travel range of $100\ \mu\text{m} \times 100\ \mu\text{m}$. This paper extends these ideas and improves system performance by proposing an automatic control switching mechanism employing model response prediction (Clarke et al. 1987). For the PZT stage, we designed two robust controllers: one provided fast but oscillatory responses and the other gave smooth but slow responses. Particle swarm optimization (PSO) algorithms (Kennedy 2010) were then applied to derive the corresponding robust PID controllers to decrease the computational loads. At each step, we estimated system responses by different controllers in the future horizon and defined penalty functions to estimate the corresponding system costs. The mechanism then switched the controllers to minimize the system cost. For the motor stage, a gain-scheduling control was applied to allow movement at a fast speed for large travel distances and at a low speed for precision positioning when the target was within $\pm 50\ \mu\text{m}$ (the stroke of the PZT stage).

We then combined the PZT stage and the motor stage to demonstrate its tracking ability for large travel distances. The system precision was significantly improved by the proposed switching control mechanism. The combined stage was also integrated with a two-photon polymerization (TPP) system for microfabrication. TPP is a technique for fabricating microstructures that is based on Two-Photon Absorption (TPA), where a molecule of material simultaneously absorbs two photons from a focused laser beam to become excited to a higher energy level (Chung et al. 2013; Lin et al. 2011). For instance, Du et al. (2016) presented hybrid TPP methods to make a 0.5 mm grating. Sun and Kawata (2003) improved the resolution of TPP and fabricated 3D microstructures. Wang et al. (2016a, 2017) integrated a long-stroke precision stage with a TPP system to fabricate linear gratings and micro-lenses with a Fresnel zone plate (FZP). The present paper applies the proposed switch control mechanism to make microstructures and compares the performance of different control methods. The proposed control scheme can effectively improve precision positioning and micro-manufacturing.

The remainder of this paper is arranged as follows: Sect. 2 describes the system and derives stage models by experiments. Section 3 designs robust controllers for the PZT stage. We then derive the corresponding robust PID controllers and propose a control switching mechanism. Section 4 describes the design of a gain scheduling controller for the motor stage. We then combine the motor stage and the PZT stage with a double-loop control structure for experimental verification. Section 5 integrates the combined stage with the TPP system for micro-lens

fabrication. The optical properties of the lenses are measured to demonstrate the effectiveness of the proposed control scheme. We then draw conclusions in Sect. 6.

2 System description and modeling

The combined stage consists of a PZT stage and a motor stage, as shown in Fig. 1a, with the specifications indicated in Table 1. The PZT stage controls displacements on the x- and y-axes, and is equipped with encoders of 1.22 nm resolution. The motor stage provides $100\ \text{mm} \times 100\ \text{mm} \times 40\ \text{mm}$ displacements on the x-, y-, and z- axes with a resolution of $0.1\ \mu\text{m}$. We integrated the PZT and the motor stage to accomplish large travel ranges with nm precision. The control structure of the combined stage is shown in Fig. 1b, where the data acquisition (DAQ) cards (PCI-6220, PCI-6221, PCI-6229, and PCIe-6323) are used to transmit the control signals and to receive the output signals. For example, for the PZT stage, the analog input voltage signals (v_{Px} and v_{Py}) are magnified tenfold by the amplifier, while the corresponding digital position signals

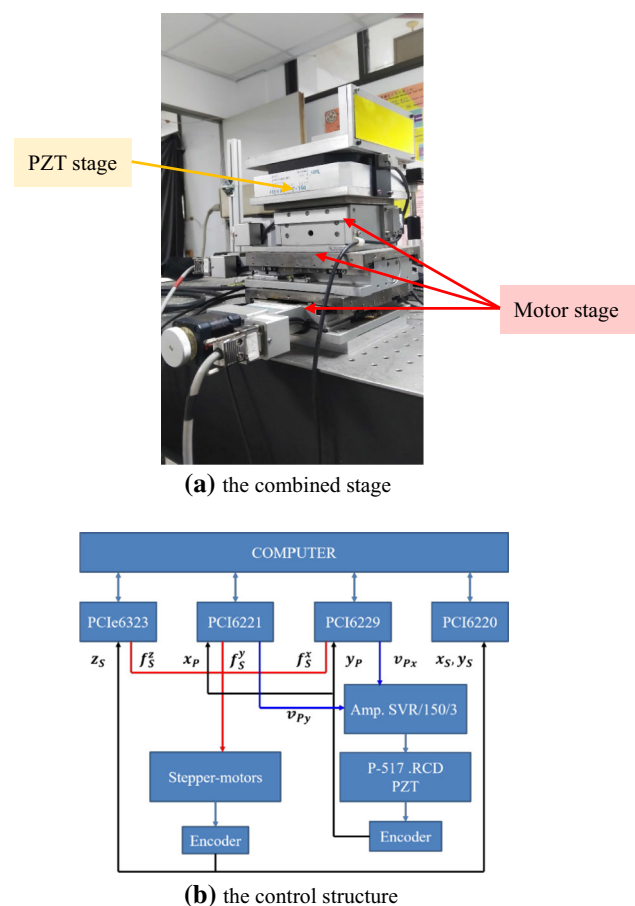


Fig. 1 The long stroke precision positioning system. **a** The combined stage, **b** the control structure

Table 1 Specifications of the combined stage

P-517.RCD PZT stage (PI Motion 2018)	
Active axis	x, y
Maximum stroke	– 50 to 50 μm
Mass	1.4 kg
SVR/150/3 amplifier (Piezomechanik GmBH 2018)	
Output voltage range	– 30 to 150 V
Max gain	30 (tunable)
Noise	0.3 mV _{pp}
ALS-510-H2P stepper (Chuo Precision 2018)	
Active axis	x, y
Maximum stroke	100 mm
Resolution	0.1 μm
Maximum loading	40kgf
Maximum command	99,999 pulse/s
ALV-104-HP stepper (Z-Lift Stages 2018)	
Active axis	Z
Maximum stroke	40 mm
Resolution	0.1 μm
Maximum loading	10 kgf
Maximum command	60,000 pulse/s

(x_P and y_P) are received through PCI-6221 and PCI-6229. For the motor stage, we input the frequency signals (f_S^x, f_S^y , and f_S^z) to control its velocities and to measure the displacement signals (x_S, y_S , and z_S).

$$G_P^{x_0} = \frac{1464s^3 + 6.794 \times 10^4 s^2 + 3.935 \times 10^8 s + 8.735 \times 10^9}{s^4 + 233.6s^3 + 2.658 \times 10^5 s^2 + 5.297 \times 10^7 s + 9.648 \times 10^8},$$

$$G_P^{y_0} = \frac{49.6s^4 + 2.911 \times 10^5 s^3 + 6.036 \times 10^8 s^2 + 1.213 \times 10^{11} s + 1.234 \times 10^{14}}{s^5 + 447.5s^4 + 9.199 \times 10^5 s^3 + 2.535 \times 10^8 s^2 + 1.448 \times 10^{11} s + 2.163 \times 10^{13}},$$

We conducted open-loop identification experiments to derive the stage models. Given the input swept sinusoidal signals with a frequency range of 0.01–100 Hz for the PZT stage and a frequency range of 0.01–20 Hz for the motor stage, we measured the corresponding output signals and applied the MATLAB command *tfest* to derive their transfer functions (Wang et al. 2017). Considering system uncertainties, we repeated the experiments ten times on each stage axis and obtained the following transfer functions:

$$G_P^{x_i}(s) = T_{v_x \rightarrow x_P}, G_P^{y_i}(s) = T_{v_y \rightarrow y_P},$$

$$G_S^{x_i}(s) = T_{f_S^x \rightarrow x_S}, G_S^{y_i}(s) = T_{f_S^y \rightarrow y_S}, i = 1, 2, \dots, 10, \quad (1)$$

where G_P^x and G_P^y (G_S^x and G_S^y) represent the transfer functions of the PZT (motor) stage on the x- and y-axes,

respectively. Because of system uncertainties and nonlinearities, the transfer functions derived from each experiment were slightly different. Therefore, we can regard the systems as linear models with uncertainties, and apply the gap metric to select the nominal plants for the controller design. Assume a plant G_0 has a left coprime factorization (LCF) (Glover and McFarlane 1989) as follows:

$$G_0 = \tilde{M}^{-1} \tilde{N},$$

where $\tilde{M}, \tilde{N} \in RH_\infty$ and $\tilde{M}\tilde{M}^* + \tilde{N}\tilde{N}^* = I$. Suppose a perturbed plant G_Δ can be represented as:

$$G_\Delta = (\tilde{M} + \Delta_{\tilde{M}})^{-1} (\tilde{N} + \Delta_{\tilde{N}}),$$

in which $\Delta_{\tilde{M}}, \Delta_{\tilde{N}} \in RH_\infty$. The gap between G_0 and G_Δ can be defined as (Georgiou and Smith 1990): The smallest value of $\|[\Delta_{\tilde{M}} \Delta_{\tilde{N}}]\|_\infty < \varepsilon$, which perturbs G_0 into G_Δ , and is denoted as $\delta(G_0, G_\Delta)$. Therefore, the nominal plant G_0 can be decided based on the following principle:

$$G_0 = \arg \left\{ \varepsilon = \min_{G_0} \max_{G_i} \delta_g(G_0, G_i) \right\}, \forall G_i.$$

That is, the selection of the nominal plants is to minimize the maximum gaps between G_0 and G_i . From the obtained models of (1), the nominal plants of the PZT stage were selected as the follows:

with system gaps of $\delta_g(G_P^{x_0}, G_P^{x_i}) \leq 0.0176$ and $\delta_g(G_P^{y_0}, G_P^{y_i}) \leq 0.0644$. Similarly, the nominal plants of the motor stage were selected as the follows:

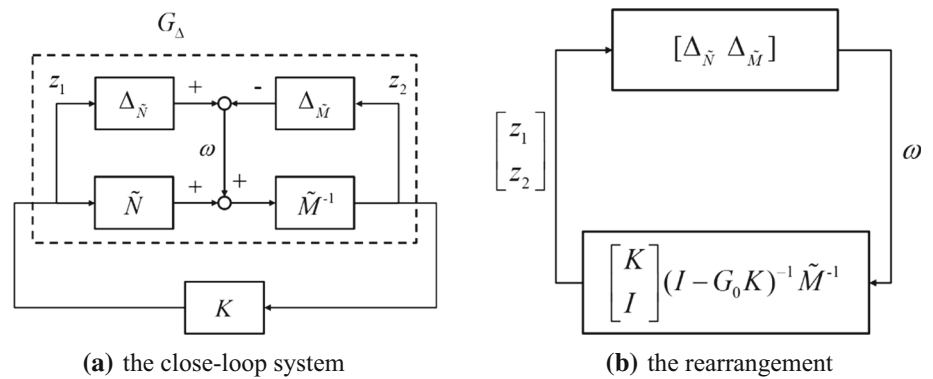
$$G_S^{x_0} = \frac{0.1006}{s}, G_S^{y_0} = \frac{0.1007}{s},$$

with the system gaps of $\delta_g(G_S^{x_0}, G_S^{x_i}) \leq 0.0048$ and $\delta_g(G_S^{y_0}, G_S^{y_i}) \leq 0.0019$. These gaps can be regarded as system variation during the operations.

3 Control design for the PZT stage

In this section, robust loop-shaping techniques are applied to design two controllers for the PZT stage: one achieved fast but oscillatory responses, while the other one gave

Fig. 2 System structure for the Small-Gain theorem. **a** The close-loop system, **b** the rearrangement



smooth but slow responses. PSO algorithms are then applied to derive the corresponding robust PID controllers to simplify hardware implementation and computation loads. Lastly, we propose an automatic control switching mechanism to combine the merits of these two controllers to achieve fast and smooth responses.

3.1 Robust loop shaping control design

A closed-loop system with a perturbed plant G_Δ and a controller K can be represented as Fig. 2a. From the Small Gain Theorem (Zhou and Doyle 1998), the system can be rearranged as Fig. 2b and the closed-loop system is internally stable for all perturbations with $\|[\Delta_M \Delta_N]\|_\infty \leq \varepsilon$ if and only if

$$\left\| \begin{bmatrix} K \\ I \end{bmatrix} (I - G_0 K)^{-1} \tilde{M}^{-1} [\tilde{M} \quad \tilde{N}] \right\|_\infty = \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I - G_0 K)^{-1} \begin{bmatrix} I & G_0 \end{bmatrix} \right\|_\infty < \frac{1}{\varepsilon}.$$

Hence, we can further define the system's stability margin as follows:

$$b(G_0, K) = \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I - G_0 K)^{-1} \begin{bmatrix} I & G_0 \end{bmatrix} \right\|_\infty^{-1},$$

so that the system is internally stable for all uncertainties if and only if $b(G_0, K) > \varepsilon$, i.e., the stability margin is greater than the system gap.

Robust loop-shaping techniques (McFarlane and Glover 1992) were applied to improve system performance. Referred to (Fig. 6a in Wang et al. 2016b), we chose weighting functions to shape the system's loop transfer functions. The selection of weighting functions was based on the following principles: increasing system gains at low-frequency range for disturbance rejection, and decreasing system gains at high-frequency for noise attenuation, and letting the magnitude slope around the crossover frequencies less steep than -40 dB/decade for system stability. Various weighting functions were designed and verified by simulation and experiments iteratively. Finally, we set the following weighting functions, which have an integral to eliminate the steady state error, and adjusted the cross-over frequencies to design a fast controller and a smooth controller:

$$W_1^x = \frac{30(s + 30\sqrt{2}\pi)}{s(s + 60\pi)}, \quad W_2^x = \frac{7.5(s + 10\sqrt{2}\pi)}{s(s + 20\pi)},$$

$$W_1^y = \frac{45(s + 30\sqrt{2}\pi)}{s(s + 60\pi)}, \quad W_2^y = \frac{12(s + 10\sqrt{2}\pi)}{s(s + 20\pi)},$$

where the subscripts $_1$ and $_2$ represent the weighting functions for the fast and the smooth controllers, respectively. The corresponding robust controllers were designed as follows:

$$K_{\infty,1}^x = \frac{1.387s^5 + 611.5s^4 + 4.334 \times 10^5 s^3 + 1.498 \times 10^8 s^2 + 1.605 \times 10^{10} s + 2.832 \times 10^{11}}{s^5 + 692.1s^4 + 3.623 \times 10^5 s^3 + 1.758 \times 10^8 s^2 + 2.14 \times 10^{10} s + 3.926 \times 10^{11}},$$

$$K_{\infty,2}^x = \frac{1.136s^5 + 338.9s^4 + 3.183 \times 10^5 s^3 + 7.969 \times 10^7 s^2 + 4.772 \times 10^9 s + 6.808 \times 10^{10}}{s^5 + 387.1s^4 + 2.884 \times 10^5 s^3 + 9.328 \times 10^7 s^2 + 5.429 \times 10^9 s + 7.735 \times 10^{10}},$$

$$K_{\infty,1}^y = \frac{1.44s^6 + 934.5s^5 + 1.449 \times 10^6 s^4 + 6.325 \times 10^8 s^3 + 2.799 \times 10^{11} s^2 + 7.295 \times 10^{13} s + 6.09 \times 10^{15}}{s^6 + 934.1s^5 + 1.227 \times 10^6 s^4 + 7.096 \times 10^8 s^3 + 2.842 \times 10^{11} s^2 + 9.335 \times 10^{13} s + 8.77 \times 10^{15}},$$

$$K_{\infty,2}^y = \frac{1.142s^6 + 583.6s^5 + 1.082 \times 10^6 s^4 + 3.561 \times 10^8 s^3 + 1.829 \times 10^{11} s^2 + 3.514 \times 10^{13} s + 1.461 \times 10^{15}}{s^6 + 610.1s^5 + 1.002 \times 10^6 s^4 + 3.993 \times 10^8 s^3 + 1.797 \times 10^{11} s^2 + 4.346 \times 10^{13} s + 1.668 \times 10^{15}}.$$

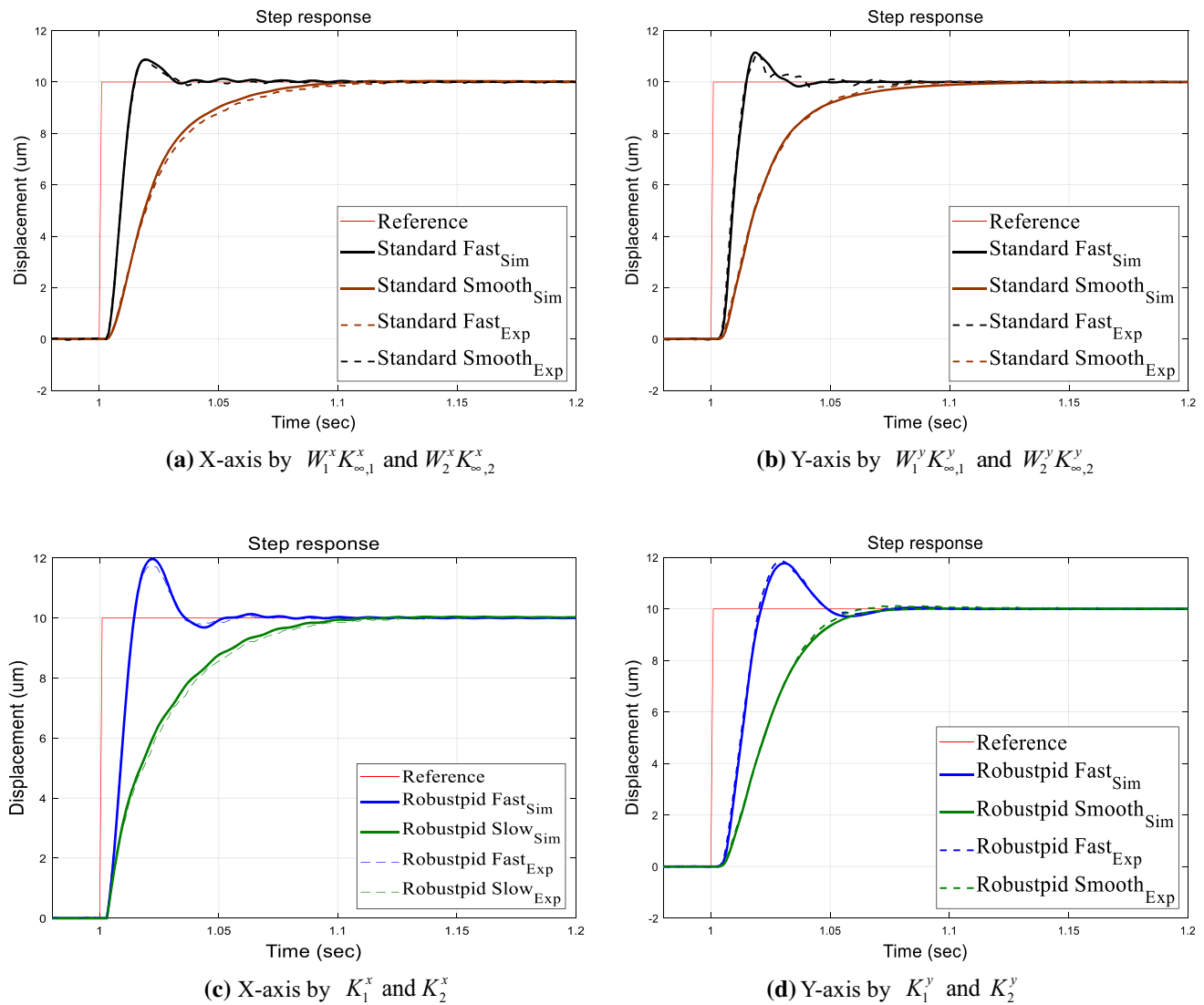


Fig. 3 Step responses of the PZT stage by robust controllers

The corresponding stability margins were as follows:

$$b(W_1^x G_P^{x_0}, K_{\infty,1}^x) = 0.5850, b(W_2^x G_P^{x_0}, K_{\infty,2}^x) = 0.6607, \\ b(W_1^y G_P^{y_0}, K_{\infty,1}^y) = 0.5704, b(W_2^y G_P^{y_0}, K_{\infty,2}^y) = 0.6589.$$

Because these stability margins are much greater than the system gaps $\delta_g(G_P^{x_0}, G_P^{x_i}) \leq 0.0176$ and $\delta_g(G_P^{y_0}, G_P^{y_i}) \leq 0.0644$, system stability can be guaranteed during operations. We implemented the weighted controllers ($W_1^x K_{\infty,1}^x$, $W_2^x K_{\infty,2}^x$, $W_1^y K_{\infty,1}^y$, and $W_2^y K_{\infty,2}^y$) for simulation and experimental verification. The results are shown in Fig. 3a, b, where experimental and simulation responses are similar. That is, the derived model can correctly predict system dynamics.

3.2 Robust PID controllers

The robust controllers can guarantee system stability and performance; however, their orders were high, which might increase the difficulties in hardware implementation and computation loadings. For this reason, PID control is preferred in industries because of its simple structure. Therefore, we combined the merits of robust control and PID control to derive the corresponding robust PID controllers based on PSO algorithms (Kennedy 2010). Consider the following PID controller:

$$K_{PID}(s) = K_P + \frac{K_I}{s} + \frac{K_D s}{T_d s + 1},$$

where $T_d = 10^{-4}$ so that the derivative term can be approximated by a proper function.

PSO is originated from the behavior of a bird flock or fish school, where each particle (bird or fish) searches its strategy that is influenced by the local best (itself) and the global best (group). At each step all particles optimize the fitness function, so that the particle swarm reaches the optimal solution after iterative movements. For the i -th particle, the velocity (V_i^{k+1}) and position (X_i^{k+1}) at the next step are determined by the following equations (Kennedy 2010):

$$V_i^{k+1} = w \times V_i^k + c_1 \times \text{rand}_1 \times (\text{Pbest}_i - X_i) + c_2 \times \text{rand}_2 \times (\text{Gbest} - X_i) \quad (2)$$

$$X_i^{k+1} = X_i^k + V_i^{k+1}, \quad (3)$$

where V_i^k and X_i^k represent the particle's current velocity and position, respectively. Pbest_i is the historical best position of the i -th particle, while Gbest is the best position among the swarm. Similarly, w is the inertia weighting function for the velocity, and c_1 and c_2 are learning factors, while rand_1 and rand_2 are random numbers between 0 and 1. Because the robust PID controller is designed to provide responses similar to those of the standard robust controller, the fitness function J is defined as follows:

$$J(K_P, K_I, K_D) = \sum_{m=1}^N \frac{\omega_m}{\Omega} \frac{J_m}{J_{m,0}} = \sum_{m=1}^N \bar{\omega}_m \bar{J}_m, \quad \Omega = \sum_{m=1}^N \omega_m, \quad (4)$$

where $J_{m,0}$ and J_m represent the m -th performance index using the standard robust controller and the robust PID, respectively, while ω_m and Ω represent the weighting for the m -th index and the summation of all weightings, respectively. That is, $\bar{J}_m$ and $\bar{\omega}_m$ are the normalized performance indexes and weightings, respectively. We considered the following performance indexes and tuned the parameters of the robust PID controller by comparing it with the standard robust controller ($W_1^x K_{\infty,1}^x$, $W_2^x K_{\infty,2}^x$, $W_1^y K_{\infty,1}^y$, and $W_2^y K_{\infty,2}^y$):

1. J_1 = the inverse of stability margin;
2. J_2 = root mean square error (RMSE) to a step command;
3. J_3 = the settling time to a step command;
4. J_4 = the rising time to a step command;
5. J_5 = the overshoot to a step command;
6. J_6 = the inverse of damping ratio of the closed loop system.

The derived robust PID controllers are as follows:

$$K_1^x(s) = 0.0335 + \frac{18.99}{s}, \quad K_2^x(s) = 0.0629 + \frac{6.25}{s},$$

$$K_1^y(s) = 1.82 \times 10^{-7} + \frac{17.58}{s}, \quad K_2^y(s) = 6.28 \times 10^{-8} + \frac{7.11}{s},$$

where the subscripts 1 and 2 indicate the fast and the smooth controllers, respectively, while the superscripts represent the x- or y-axis. We implemented the designed robust PID controllers for simulation and experiment verification. The results are shown in Fig. 3c, d, which show that the experimental and simulation responses were similar. Furthermore, the robust PID controllers (see Fig. 3c, d) had similar responses to the standard robust controllers (see Fig. 3a, b). Though the PID controllers had larger overshoots and longer settling time than the standard robust controllers, they can provide similar responses with a much simpler structure. Therefore, we applied these robust PID controllers to develop the control switching algorithms.

3.3 Controller switching by model-based prediction

As shown in Fig. 3, the fast controllers (K_1^x and K_1^y) achieved shorter rise time and settling times when compared to the smooth controllers (K_2^x and K_2^y), whereas the smooth controllers had smaller overshoots when compared to the fast controllers. Therefore, we combined the virtues of these two controllers by developing an automatic controller switching mechanism based on model response prediction. The structure is shown in Fig. 4, where the optimizer predicts the system responses by different controllers in the future horizon and chooses a controller to minimize the penalty functions.

For hardware implementation, the continuous-time models were discretized by zero-order hold, because the control signals were assumed constant between the samples. Consider the following discrete-time model:

$$A(z^{-1})y(k) = B(z^{-1})u(k-1),$$

where A and B are the denominator and numerator of the nominal plant:

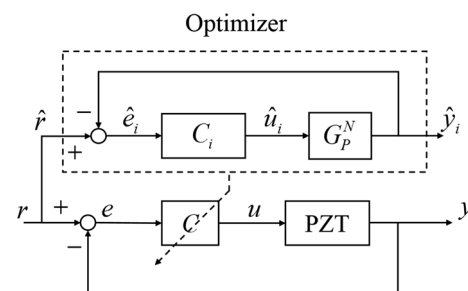


Fig. 4 Controller switching mechanism

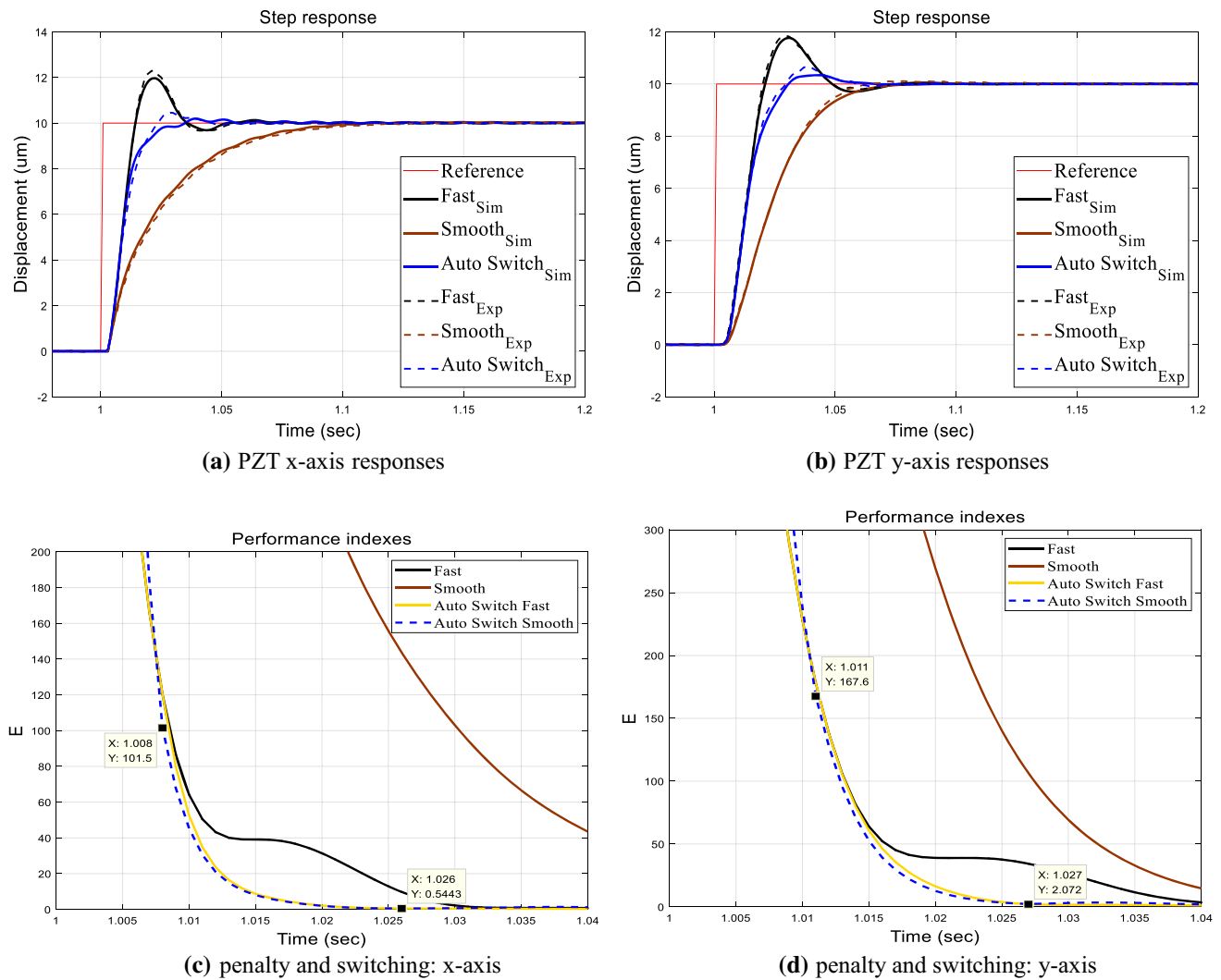


Fig. 5 Step responses for robust PID and switching mechanism. **a** PZT x-axis responses, **b** PZT y-axis responses, **c** penalty and switching: x-axis, **d** penalty and switching: y-axis

$$A(z^{-1}) = 1 + a_1 z^{-1} + \dots + a_{na} z^{-na}$$

$$B(z^{-1}) = b_0 + b_1 z^{-1} + \dots + b_{nb} z^{-nb}$$

To predict the output signal at the j -th step in the future, we consider the following unit matrix:

$$1 = E_j(z^{-1})A(z^{-1}) + z^{-j}F_j(z^{-1}) \quad (5)$$

where $E_j = 1$ and $F_j = (1 - A)z^j$. Therefore, the relationship between output y and control u can be represented as:

$$y(k) = (1 - A)y(k) + Bu(k - 1) \quad (6)$$

Multiplying (6) by z^j and substituting $F_j = (1 - A)z^j$ into (6) gives:

$$y(k + j) = F_j y(k) + Bu(k + j - 1). \quad (7)$$

Similarly, the designed controllers ($K_1^x(s)$, $K_2^x(s)$, $K_1^y(s)$, and $K_2^y(s)$) were transferred to the z -domain by bilinear (Tustin) transformation for good matching in frequency domain. Consider the i -th controller:

$$C_i(z^{-1})u(k) = D_i(z^{-1})e(k), \quad (8)$$

where C_i and D_i are the denominator and numerator, respectively, of the i -th controller:

$$C_i(z^{-1}) = 1 + c_{1,i}z^{-1} + \dots + c_{nc,i}z^{-nc}$$

$$D_i(z^{-1}) = d_{0,i} + d_{1,i}z^{-1} + \dots + d_{nd,i}z^{-nd}$$

Multiplying (8) by $C_i^{-1}z^{j-1}$ results in:

$$u(k + j - 1) = C_i^{-1}D_i e(k + j - 1) \quad (9)$$

Furthermore, substituting (9) into (7) gives:

Table 2 Step responses of the robust PID controllers and switching mechanism

	K_1^x	K_2^x	Switching control
PZT x-axis			
Simulation			
Rising time (s)	0.013	0.056	0.017
Settling time (s)	0.048	0.086	0.025
Overshoot	19.71%	0.42%	1.97%
Experiment			
Rising time (s)	0.012	0.057	0.018
Settling time (s)	0.048	0.093	0.033
Overshoot	20.77%	0.41%	2.69%
	K_1^y	K_2^y	Switching control
PZT y-axis			
Simulation			
Rising time (s)	0.019	0.046	0.023
Settling time (s)	0.066	0.062	0.050
Overshoot	17.67%	< 0.1%	3.39%
Experiment			
Rising time (s)	0.018	0.043	0.022
Settling time (s)	0.061	0.055	0.050
Overshoot	18.41%	1.11%	6.69%

$$y(k+j) = F_j y(k) + BC_i^{-1} D_i e(k+j-1).$$

Therefore, we can predict the future responses using the nominal plant and the i -th controller, as in the following:

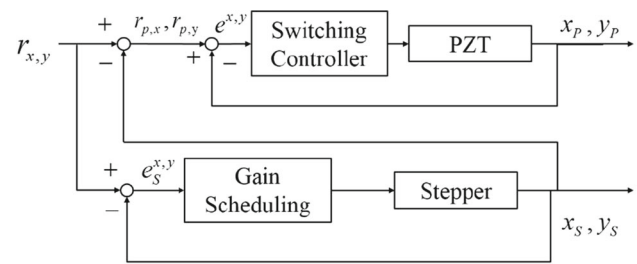
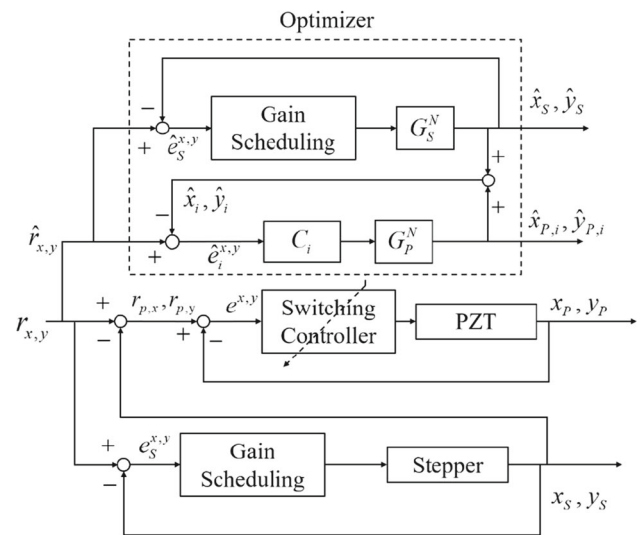
$$\hat{y}_i(k+j|k) = F_j y(k) + G_{j,i} \hat{e}(k+j-1|k),$$

in which $G_{j,i}(z^{-1}) = BC_i^{-1} D_i$ and $\hat{e}(k+j|k) = \hat{r}(k+j) - \hat{y}(k+j|k)$.

Suppose the prediction horizon was Np . We could then calculate the future output $\hat{y}(k+j)$, $j = 1, 2, \dots, Np$ and define the penalty function as follows:

$$E(Np) = \sum_{j=1}^{Np} (r(k+j) - y(k+j))^2,$$

which represents the accumulated error square in the prediction horizon. At each step, the optimizer calculated the penalty for each controller and chose one that minimized the penalty in the future horizon. We implemented the control switching mechanism for simulation and experimental verification. For example, setting $Np = 40$ with a step reference introduced at $t = 1$ s gave the step responses shown in Fig. 5, with the statistical data of Table 2. First, the switching control simultaneously combined the advantages of the fast and the smooth controllers: the rise time was similar to the fast controllers and the overshoots were close to the smooth controllers, while the settling time was better than both controllers. Second, the mechanism

**(a)** Double-loop control**(b)** Controller switching mechanism**Fig. 6** Control and estimation of the combined stage. **a** Double-loop control, **b** controller switching mechanism

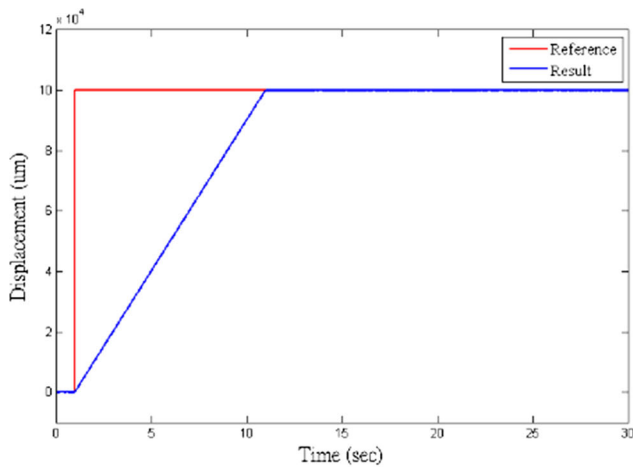
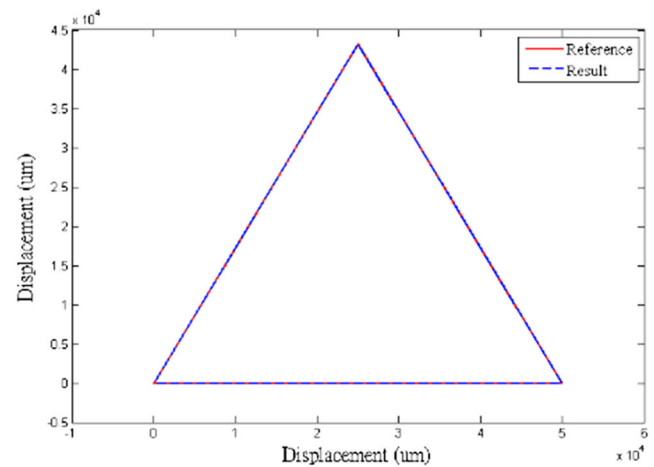
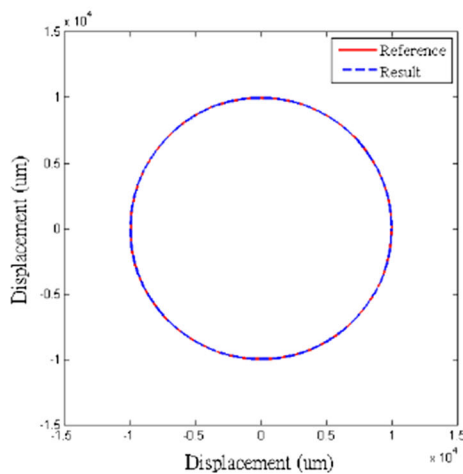
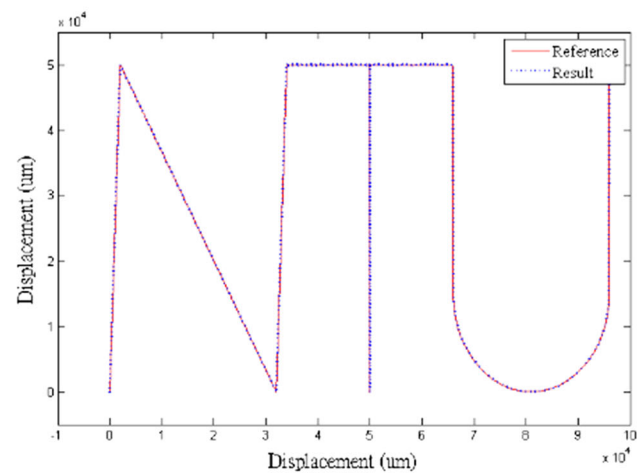
evaluated the penalty and chose a better controller at each step, as shown in Fig. 5c, d. For example, the fast control was chosen to shorten the rise time at $t = 1$ s, and then the smooth control was selected to avoid large overshoots at $t = 1.008$ s and $t = 1.011$ s for the x- and y-axis, respectively. Finally, the mechanism adopted the fast control for quick settling at $t = 1.026$ s and $t = 1.027$ s for the x- and y-axis, respectively. Consequently, the switching mechanism provided better settling time with acceptable overshoots.

4 Stage integration and control

We combined the PZT stage and the motor stage to achieve long-stroke precision positioning. A double-loop control structure was designed for the combined stage, as shown in Fig. 6a, where the errors of motor stage ($e_s^{x,y}$) were used to control the motor stage for long travel distances, while the combined errors ($e^{x,y}$) were used to regulate the PZT stage for precision positioning. The prediction model was also modified for the combined control, as shown in Fig. 6b,

Table 3 Comparison of the experimental responses of the motor stage

Steps	Properties	Gain scheduling	Robust control (Wang et al. 2016b)
100 μm	Settling time (s)	0.07	0.19
	Overshoot	< 0.1%	< 0.1%
1 mm	Settling time (s)	0.108	0.23
	Overshoot	< 0.1%	< 0.1%
10 mm	Settling time (s)	0.985	1.25
	Overshoot	< 0.1%	< 0.1%
100 mm	Settling time (s)	9.659	12.24
	Overshoot	< 0.1%	< 0.1%

**(a)** 10cm-step tracking**(b)** triangular tracking**(c)** circular tracking**(d)** word tracking**Fig. 7** Tracking responses of the combined stage. **a** 10 cm-step tracking, **b** triangular tracking, **c** circular tracking, **d** word tracking

where the following estimated outputs of the combined stage $\hat{x}_i$ and $\hat{y}_i$ were applied to decide the switching time:

$$\begin{aligned}\hat{x}_i(k+j|k) &= \hat{x}_{P,i}(k+j|k) + \hat{x}_S(k+j|k), \\ \hat{y}_i(k+j|k) &= \hat{y}_{P,i}(k+j|k) + \hat{y}_S(k+j|k).\end{aligned}$$

Note that the predictions of the motor stage ($\hat{x}_S$ and $\hat{y}_S$) were similar to the predictions for the PZT stage ($\hat{x}_{P,i}$ and $\hat{y}_{P,i}$) described in Sect. 3.

Table 4 Tracking responses of the combined stage

Reference	Proposed control		Control in Wang et al. (2017)	
	RMSE (nm)	Settling time (s)	RMSE (nm)	Settling time (s)
10-cm step	1. 1.2272	1. 9.793	1. 1.8417	1. 9.791
	2. 1.1131	2. 9.793	2. 1.8795	2. 9.778
	3. 1.0437	3. 9.789	3. 2.2518	3. 9.794
Reference	RMSE (nm)	MAE (μm)	RMSE (nm)	MAE (μm)
5 × 5×5 cm triangle				
X-axis	1. 193.76	1. 0.5900	1. 481.54	1. 1.7020
	2. 203.00	2. 0.7520	2. 474.33	2. 1.7140
	3. 200.40	3. 0.6550	3. 475.16	3. 1.7590
Y-axis	1. 381.32	1. 1.5158	1. 565.34	1. 3.2285
	2. 390.19	2. 1.3410	2. 531.07	2. 2.8347
	3. 381.47	3. 1.3485	3. 523.14	3. 2.8500
1 cm radius circle				
X-axis	1. 253.39	1. 0.8623	1. 573.25	1. 2.1310
	2. 327.18	2. 0.8630	2. 561.29	2. 2.4382
	3. 266.04	3. 0.7690	3. 525.28	3. 2.9328
Y-axis	1. 480.33	1. 1.8255	1. 699.96	1. 3.1486
	2. 486.09	2. 2.1733	2. 741.66	2. 3.4688
	3. 457.41	3. 1.6900	3. 730.11	3. 3.7246
NTU				
X-axis	1. 141.73	1. 0.9884	1. 328.13	1. 2.8161
	2. 162.14	2. 1.0247	2. 333.69	2. 3.0618
	3. 164.41	3. 0.9753	3. 350.06	3. 3.1293
Y-axis	1. 408.05	1. 4.4622	1. 666.54	1. 5.3987
	2. 505.83	2. 3.7390	2. 670.06	2. 4.9947
	3. 458.01	3. 4.2772	3. 4063.05	3. 4.5741

4.1 Stepper control

In our previous work (Wang et al. 2016b), we designed robust controllers for the motor stage. However, because the references of the PZT stage were influenced by the position of the motor stage, the speed of the motor stage should be reduced when the errors $e_S^{x,y}$ are within the PZT travel range ($\pm 50 \mu\text{m}$) to achieve smooth responses. Therefore, we applied the following gain-scheduling control to the motor stage:

$$k_S^{x,y} = k, \begin{cases} k = 399.996, |e_S^{x,y}| \leq 10 \\ k = 39.9996 \times |e_S^{x,y}|, 10 < |e_S^{x,y}| < 50, \\ k = 1999.98, |e_S^{x,y}| \geq 50 \end{cases}$$

where the maximum gain was set as $k = 1999.98$ by considering the motor's maximum command (99,999 pulse/s, see Table 1), so that the command was $1999.98 \times 50 = 99999$ when $e_S^{x,y} = \pm 50 \mu\text{m}$. On the other hand, the minimum gain ($k = 399.996$) was chosen to give smooth

responses, based on simulation and experimental results. Furthermore, the following saturation function was set to avoid desynchronizing when making large-distance movements:

$$\frac{u(k) - u(k-1)}{T} \leq 9999900 \text{ (pulse/sec}^2\text{)},$$

where $u = k_S^{x,y} \cdot e_S^{x,y}$ with a sampling time of $T = 0.001$ s. We compared the step responses of the motor stage using the proposed gain-scheduling control and the robust control designed in (Wang et al. 2016b). The statistical data for the experiments are shown in Table 3, where the proposed scheduling control achieved better performance.

4.2 Combined stage control

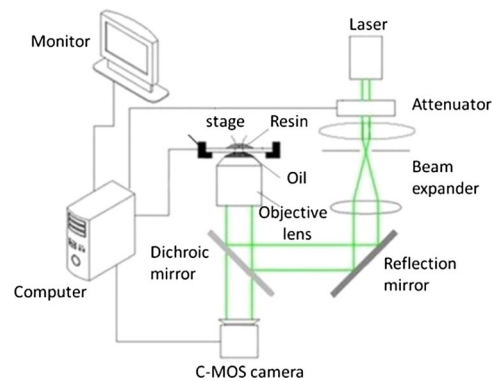
The designed control structure (see Fig. 6) was implemented with the combined stage to conduct the following four tracking experiments for experimental verification: 10 cm steps, 5 cm × 5 cm × 5 cm triangles, 1 cm circles,

Fig. 8 The system integration for micro-fabrication.

a Combined stage and TPP system, **b** the TPP module (Chung et al. 2013)



(a) Combined stage and TPP system



(b) The TPP module (Chung et al. 2013)

and a word “NTU” as shown in Fig. 7. To demonstrate the virtues of the proposed control, we compared the system responses using the proposed control and the control in Wang et al. (2017), where integral control with gain scheduling was applied to the PZT stage, while robust control with a feedforward compensator was applied to the motor stage. Each experiment was conducted three times with the results are illustrated in Table 4. First, the average RMSE for the step responses was reduced from 1.991 nm to 1.128 nm, with similar settling time. Second, for the triangular tracking the average RMSE was greatly reduced from 508.43 nm to 291.69 nm (43% improvement), while the max absolute error (MAE) was significantly decreased from 2.348 to 1.034 μm (54% improvement) using the proposed control. Third, the average RMSE (MAE) for the circular tracking was greatly reduced from 638.592 (2.974 μm) to 378.407 nm (1.364 μm), with 41% (54%) improvement. Lastly, the average RMSE (MAE) for the word tracking was significantly improved from 1068.59 (3.996 μm) to 306.70 nm (2.578 μm) by the proposed control. That is, the proposed control structure and switching mechanism were effective in improving the tracking performance of the long-stroke precision stage.

Table 5 Parameters of the TPP system

Laser		
Laser power		4mW
Exposure time		1 ms
OLYMPUS UPlanFL N Objective		
Magnification		100x
Numerical aperture (NA)		1.30
Material		
Resin		Ormocomp
Initiator		1.2% photo initiator

5 Micro-lens fabrication by the combined stage

The combined stage was further integrated with a TPP (Chung et al. 2013; Wu et al. 2009) to fabricate micro-lenses by FZP (Wang et al. 2017), as shown in Fig. 8. TPP is based on the principles of TPA, where a molecule of material simultaneously absorbs the energy of two photons to become excited to a higher energy level (Chung et al. 2013; Lin et al. 2011). We used an Nd:Yag laser to trigger

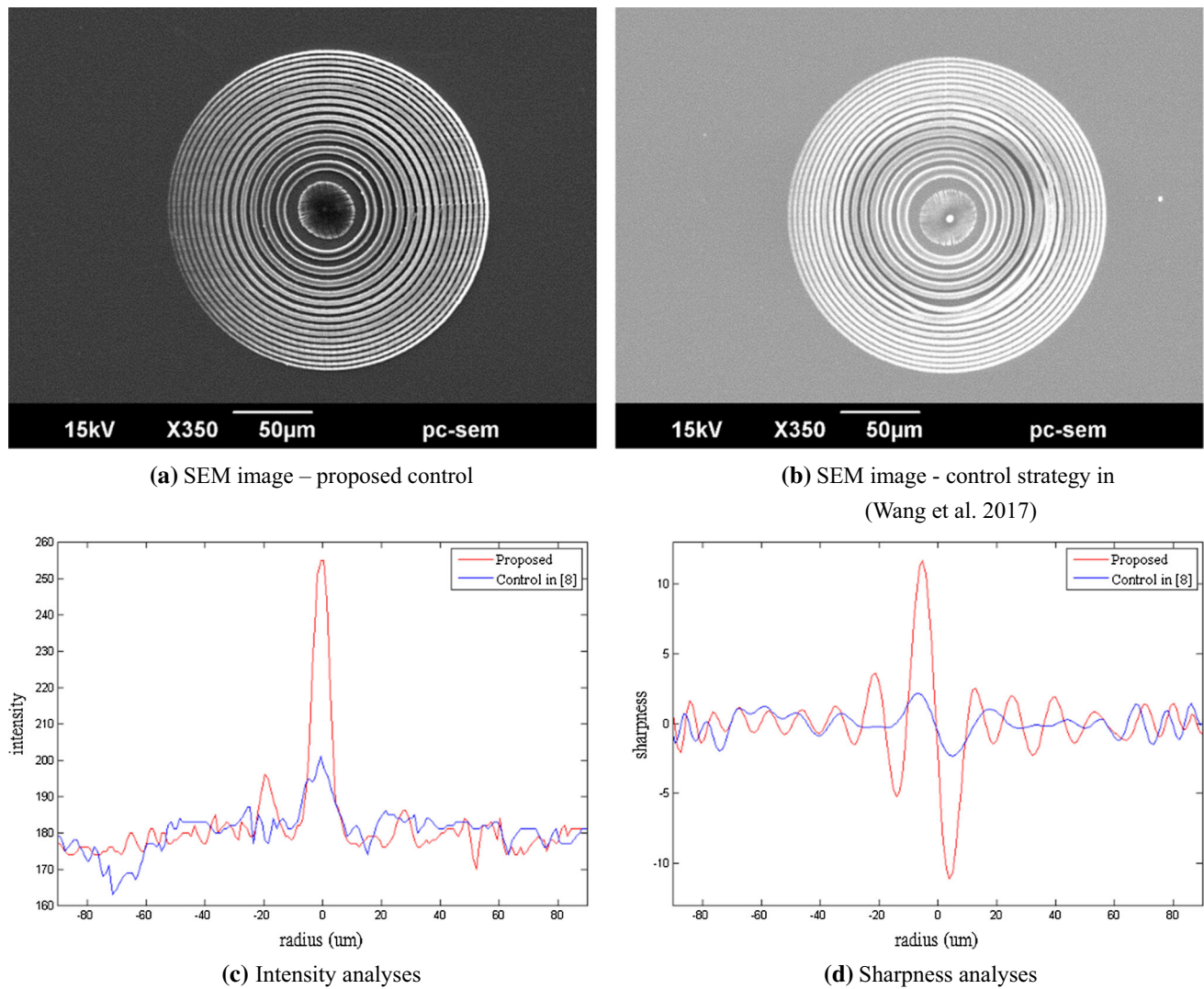


Fig. 9 Optical properties of the FZP. **a** SEM image-proposed control, **b** SEM image-control strategy in Wang et al. (2017), **c** intensity analyses, **d** sharpness analyses

the TPP of resin to make micro-structures. As shown in Fig. 8(b), the laser was filtered by the attenuator and passed through the beam expander, and the laser was reflected by the dichroic mirror at a specific wavelength and focused on the resin by the objective lens (Chung et al. 2013). Because the TPP processes can be influenced by several factors, such as materials and laser power, the experiments were conducted under the same conditions, as illustrated in Table 5. We prepared the resin in advance and placed it on the adaptor attached to the stage. The laser was fixed onto a particular point and the micro-lens was fabricated by moving the combined stage, so that precision positioning of the stage would greatly influence the qualities of the micro-lens.

The FZP is a micro-lens that consists of radially symmetric rings that focus light by diffraction (Tamura 2011). We designed an FZP with a radius of 99.4 μm to focus red

light with a wavelength of 632.8 nm at an ideal focal length of 500 μm (Wang et al. 2016a). The light spot from a FZP at the ideal focal length was then captured by a Complementary Metal–Oxide–Semiconductor (CMOS) camera, as shown in Fig. 9a. Therefore, we could evaluate the optical quality of the lens by examining the intensity and sharpness of the CMOS signals, as shown in Fig. 9b. The intensity is defined as the brightness of the image. Because the digital images consisted of the RGB (red, green, and blue) colors that were numerically represented as binary numbers from 0 to 255, the intensity can be calculated by the following equation (ITU 2017):

$$Y(\text{intensity}) = 0.299 \times R(\text{red}) + 0.587 \times G(\text{green}) + 0.114 \times B(\text{blue}),$$

Table 6 TPP experiments using the combined stage

	Control in Wang et al. (2017)		Proposed control	
	X-axis	Y-axis	X-axis	Y-axis
RMSE (μm)	0.1237	0.1552	0.0742	0.0880
MAE (μm)	1.1404	1.9680	0.6226	1.3957

where R:G:B = 1:1:1 because a white light source was applied, i.e. $Y = R = G = B$. Conversely, the sharpness of the signals is defined as the derivative of the intensity.

The combined stage was controlled by the proposed method to fabricate the micro-lenses. For comparison, we also implemented the control strategies described in (Wang et al. 2017). The results are shown in Fig. 9 and Table 6. First, the FZP made by the proposed stage control (see Fig. 9a) was better than the FZP made by the controllers designed in (Wang et al. 2017) (see Fig. 9b) because the RMSE and MAE were significantly reduced (see Table 6). Second, the optical qualities of the micro-lenses were also greatly improved by the proposed method, as shown in Fig. 9c, d). That is, the proposed control structure and mechanism were effective in improving the precision positioning and manufacturing.

6 Conclusion

This paper has proposed switching control strategies for a combined stage, which consisted of a PZT stage and a motor stage, for improving micro-lens fabrication. For the PZT stage, two robust controllers were designed that individually achieve fast transient responses and smooth steady responses. We then developed a control switching mechanism based on model response prediction that allowed simultaneous achievement of fast and smooth responses. For the motor stage, a gain-scheduling control was designed to let the stage move at the maximum speed for long travel distances and at a reduced speed when the positioning errors were within the PZT stage travel ranges. We combined the PZT and motor stage with a double-loop control structure to achieve a 10 cm travel distance with a precision of 1.23 nm. The combined stage was also integrated with a TPP system to fabricate micro-lenses with a radius of 99.4 μm by FZP. Based on the optical properties of these lenses, the proposed control is deemed effective in improving precision positioning and micro manufacturing.

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